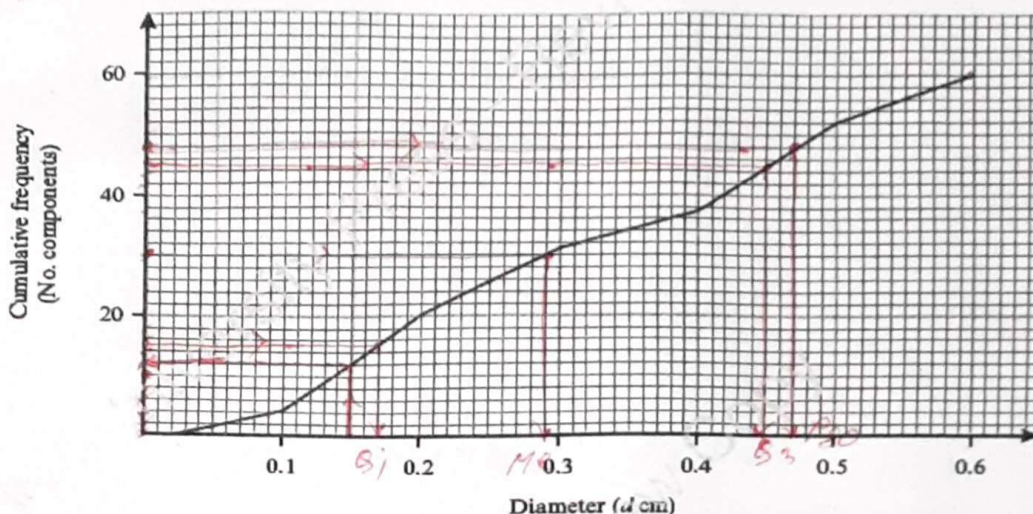




United International University
School of Science and Engineering
Department of Computer Science and Engineering
 Mid Term Examination || Trimester: Fall – 2025
 Course: Math -2205 (Probability & Statistics)
 Total marks – 30 || Duration – 1 hour 30 minutes

[Note that the number of marks is given in brackets [] at the end of each question or part question. You are requested to answer all the questions in order.]

- Q1** (a) The diameters, d cm of 60 cylindrical electric components, are represented in the following cumulative frequency graph



- (i) Find the median and inter-quartile range of the diameter of the cylindrical electric components. [3]
 $M2 = 0.27$ $IBR = 0.45 - 0.17 = 0.28$ [2]
 (ii) Showing necessary working, state the nature of the distribution [2]
 $S1 \sim MC = 0.12 < S3 \sim MC = 0.16$ Right-skewed
 (b) Estimate the number of components that have a diameter less than 0.15 cm [1]
 12 (From graph)
 (c) Estimate the value of K , given that 20% of the components have diameter of K cm or more [2]
 Above 20% $\rightarrow$ Below 80% $\rightarrow P_{80} = 0.47$

- Q2** (a) The following data set represents the scores obtained by 12 students in a mathematics test:

45, 48, 50, 52, 53, 55, 56, 58, 60, 62, 95, 97

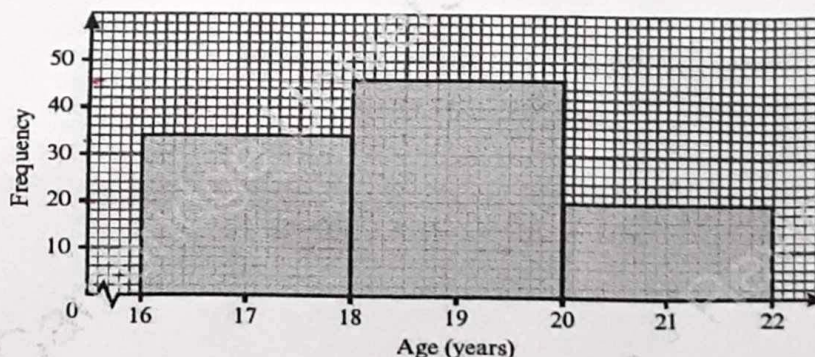
$M2 = 55.5$, $S1 = 51$, $S3 = 61$
 $IBR = 61 - 51 = 10$

Using an appropriate statistical method, to determine whether there exists any possible outlier in the data set. [3]

outliers

$LL = 51 - 1.5 \times 10 = 36$
 $UL = 61 + 1.5 \times 10 = 76$

- (b) The ages, in years of some children are represented in the following histogram



Construct a frequency distribution based on the above histogram and hence find the mean age of the children. [4]

class	x	f	fx
16-18	17	34	578
18-20	19	46	874
20-22	21	20	420
		100	1872

P.T.O.

$\bar{x} = \frac{1872}{100}$
 $= 18.72$

- Q3 (a) Assume that the data given in the following table represents the GPA distribution of 40 students of UIU.

GPA	0 - 1	1 - 2	2 - 3	3 - 4
Number of students	6	$f=7$	18	9

- (a) Find the value of f . $f = 40 - (6 + 18 + 9) = 7$ [1]
 (b) Estimate the mode of the distribution. $Mo = 2 + \frac{11}{11+9} \times 1 = 2.55$ [3]
 (c) Calculate the percentage of students who will not receive a penalty, given that students with a GPA less than 1.00 will receive a penalty. $\% \rightarrow \frac{7+18+9}{40} \times 100\% = 85\%$ [1]

(b) Two sections of a university course, Section A and Section B, recorded the following statistics for the number of hours students spend studying per week:

Section	Mean study hours ($\bar{x}$)	Standard deviation (σ)	$cv(\%)$
A	25 hours	5 hours	$\frac{5}{25} \times 100\% = 20\%$
B	30 hours	9 hours	$\frac{9}{30} \times 100\% = 30\%$

Calculate the coefficient of variation (CV) for both sections and hence identify which section shows greater consistency in study hours. $cv(A) < cv(B)$, A has greater consistency. [3]

- Q4 (a) What kind of correlation exists between relations:

- (i) daily temperature and ice-cream sales $\rightarrow$ positive correlation [2]
 (ii) age of a car and its resale value $\rightarrow$ negative correlation

- (b) The ranks of 6 delivery agents based on delivery speed and customer rating are given: [3]

Agent	A	B	C	D	E	F
Speed Rank	1	2	3	4	5	6
Rating Rank	2	1	4	3	6	5

Calculate Spearman's rank correlation coefficient and interpret the result.

- (c) A logistics company studies the relationship between distance traveled (x , in km) and fuel consumption (y , in liters) for its delivery vehicles. Using past data, the least squares regression line of fuel consumption on distance is obtained as: [2]

$$y = 2.5 + 0.08x$$

- (i) Interpret the **slope** of the regression line in the context of this problem
 (ii) Estimate the fuel consumption for a trip of 50 km using the given regression line

(i) $\checkmark$ will increase/decrease by 0.08 for the increase/decrease of x by 1.
 (ii) $y(50) = 6.5 \text{ L.}$

Formulae:	For the regression line: $y = a + bx$	$a = \frac{\sum y \sum x^2 - \sum x \sum xy}{n \sum x^2 - (\sum x)^2}$ and $b = \frac{n \sum xy - \sum x \sum y}{n \sum x^2 - (\sum x)^2}$
	Coefficient of correlation	$r = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sqrt{\sum (x - \bar{x})^2 \sum (y - \bar{y})^2}}$
	Spearman's rank correlation coefficient:	$\rho = 1 - \frac{6 \sum d_i^2}{n(n^2 - 1)}$